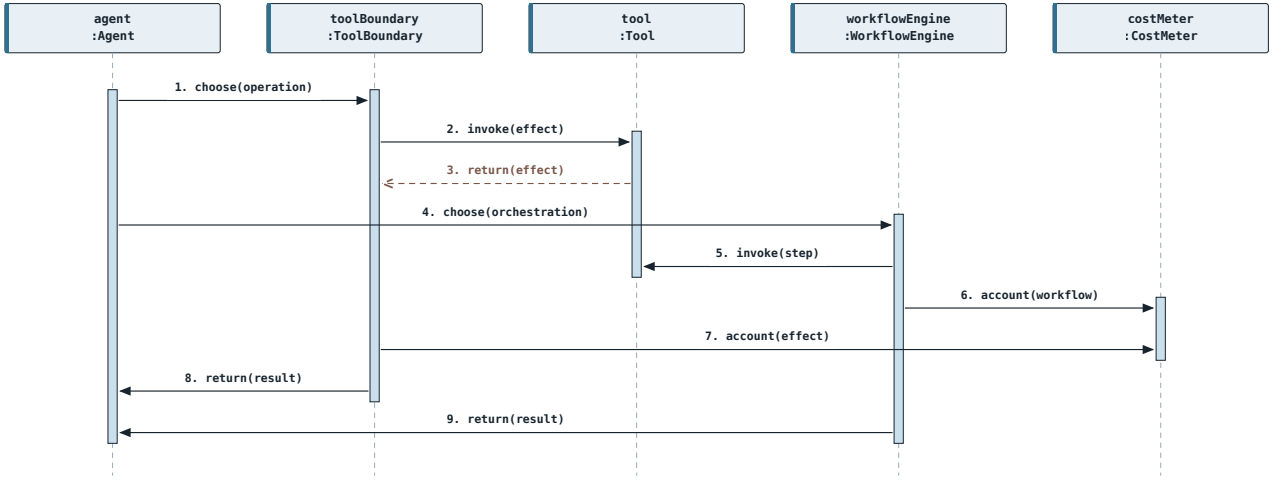
P11 Tool-workflow boundary

Primitive / OBJECT INTERACTION / ANALYTICAL COLLABORATION

Distinguish an operation from an orchestration.

OBJECT INTERACTION

solid arrow: call/message · dashed open arrow: return/failure · analytical order, not an observed trace



POINTCUT / PATTERN INTERFACE	
CONTEXT	The harness exposes both primitive effects and multi-step procedures.
TRIGGER	The agent chooses a tool or workflow entry point.
STRUCTURE	Separate effect/result contracts and workflow ownership of sequencing.
EFFECT	Workflow orchestration is not mistaken for a single external side effect.

CONTRACT / REQUIRED BEHAVIOR	
REQUIRES	Tool and workflow descriptors have distinct schemas and execution paths.
ENSURES	Retry, budget, and result accounting bind to the correct unit.
INVARIANT	A workflow cannot hide duplicate irreversible tool effects.
VIOLATION	Nested failures and retries produce explicit ownership and cost records.

Sources: protocol/pattern_diagram_models.yaml (participants + messages); protocol/harness_pattern_specification.yaml (pointcut + contract).
Analytical collaboration only; no repository execution or message order is asserted.